

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – IV
PAPER –1
TEST DATE: 18-01-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 –10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

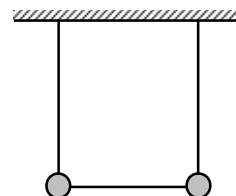
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

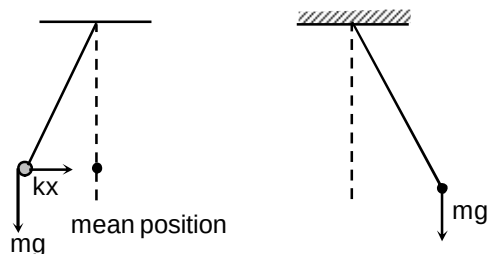
1. Two identical simple pendulums each of length ℓ are suspended from the ceiling. Their bobs each of mass m are connected by a very light relaxed elastic cord of force constant k as shown in the figure. If the bobs are symmetrically pulled apart slightly and released, what is period of the ensuing oscillations?



- (A) $\pi \left(\frac{2k}{m} + \frac{g}{\ell} \right)^{-\frac{1}{2}}$
- (B) $\pi \left\{ \left(\frac{2k}{m} \right)^{-\frac{1}{2}} + \left(\frac{g}{\ell} \right)^{-\frac{1}{2}} \right\}$
- (C) $\pi \left\{ \left(\frac{k}{m} + \frac{g}{\ell} \right)^{-\frac{1}{2}} + \left(\frac{g}{\ell} \right)^{-\frac{1}{2}} \right\}$
- (D) $\pi \left\{ \left(\frac{2k}{m} + \frac{g}{\ell} \right)^{-\frac{1}{2}} + \left(\frac{g}{\ell} \right)^{-\frac{1}{2}} \right\}$

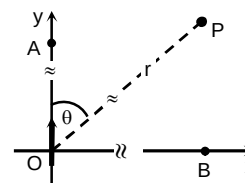
Ans. D

Sol.



Add time period of left and right motion from mean position.

2. In the given figure, dipole moment $\vec{m}$ of a small magnetic dipole placed at the origin is shown by a small directed line segment. Magnetic induction vectors at distant points A (0, y) on the y-axis and B (x, 0) on the x-axis respectively are



- (A) $\frac{\mu_0 m}{4\pi y^3} \hat{j}$ and $\frac{\mu_0 m}{2\pi x^3} \hat{i}$

(B) $\frac{\mu_0 m}{2\pi y^3} \hat{j}$ and $\frac{\mu_0 m}{4\pi x^3} \hat{i}$

(C) $\frac{\mu_0 m}{4\pi y^3} \hat{j}$ and $-\frac{\mu_0 m}{2\pi x^3} \hat{j}$

(D) $\frac{\mu_0 m}{2\pi y^3} \hat{j}$ and $-\frac{\mu_0 m}{4\pi x^3} \hat{j}$

Ans. D

Sol. Use result M.F. due to dipole on axis $= \frac{\mu_0 \vec{m}}{2\pi r^3}$

Use result M.F. due to dipole on equatorial line $= -\frac{\mu_0 \vec{m}}{4\pi r^3}$

3. Three small balls of the same mass, white (w), green (g), and blue (b), are fixed by weightless rods at the vertices of the equilateral triangle with side l . The system of balls is placed on a smooth horizontal surface and set in rotation about the centre of mass with period T . At a certain instant, the blue ball tears away from the frame. Determine the distance L between the blue and the green ball after the time T .

(A) $\sqrt{\frac{2}{3} + \left(\pi\sqrt{3} - \frac{1}{2}\right)^2}$

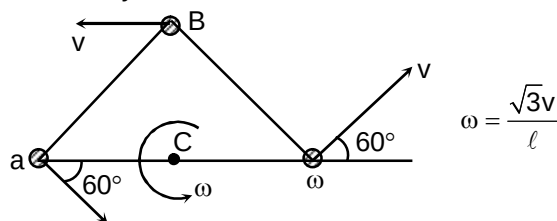
(B) $\sqrt{\frac{4}{3} + \left(\pi\sqrt{3} - \frac{1}{2}\right)^2}$

(C) $\sqrt{\frac{4}{3} + (\pi\sqrt{3} - 1)^2}$

(D) $\sqrt{\frac{3}{4} + \left(\pi\sqrt{3} - \frac{1}{2}\right)^2}$

Ans. D

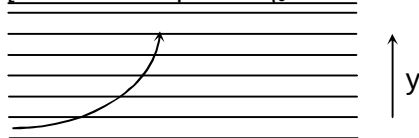
Sol. Displacement of G = Displacement w.r.t. C + Displacement of C
Further, by mathematical calculations, it can be solved.



4. The refractive index of the medium within a certain region, $x > 0, y > 0$, changes with y . A thin light ray travelling in the x -direction strikes the medium at right angles and moves through the medium along a circular arc.

How does the refractive index depend on y ?

[R : Radius of path & η_0 : Initial refractive index]



(A) $\frac{2R}{R-y} \eta_0$

(B) $\frac{R}{R-y} \frac{\eta_0}{2}$

(C) $\frac{R+y}{R} \eta_0$

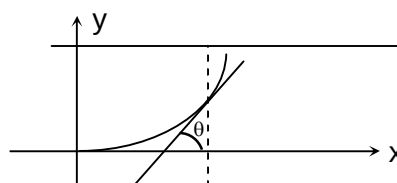
(D) $\frac{R}{R-y} \eta_0$

Ans. D

Sol. Equation of circle is
 $(x-0)^2 + (y-R)^2 = R^2$... (1)

$\frac{dy}{dx} = \tan \theta$... (2)

By Snell's law $\sin(90 - \theta) \times \eta = 1 \times \eta_0$... (3)



Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A closed organ pipe of length ℓ is sounded together with another closed organ pipe of length $\ell + x$ ($x \ll \ell$). Both vibrate in their fundamental modes. If the speed of sound is v , then the beat frequency heard is

(A) $\frac{vx}{2\ell^2}$

(B) $\frac{vx}{4\ell^2}$

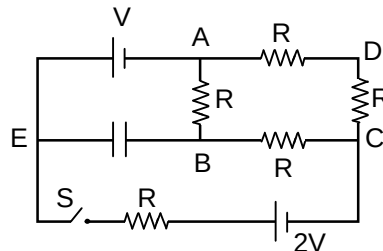
(C) $\frac{vx^2}{4\ell}$

(D) None of these

Ans. B

Sol.
$$\Delta f = \frac{v}{4\ell} - \frac{v}{4(\ell+x)} = \frac{v}{4\ell} \left[1 - \frac{1}{\left(1 + \frac{x}{\ell}\right)} \right] = \frac{v}{4\ell} \left[1 - 1 + \frac{x}{\ell} \right] = \frac{vx}{4\ell^2}$$

6. Five equal resistances, a capacitor and two batteries are connected in a circuit as shown in the figure. The capacitor is initially uncharged and each battery has negligible internal resistance. It is given that $R = 1\Omega$, $C = 1\mu\text{F}$ and $V = 8$ volt. On the basis of above information answer the following questions, if switch 'S' is closed at $t = 0$. (Assume $e^{-1} = 0.37$, if required)



- (A) Current in the branch of capacitor at $t = 5/8 \mu\text{s}$ is (approximately) 6 A
 (B) Current in the branch AB at $t = 5/8 \mu\text{s}$ is (approximately) 1.7 A
 (C) Potential difference between points A and C at $t = 5/8 \mu\text{s}$ is (approximately) 2.5 V
 (D) Potential difference between points A and C at $t = 5/8 \mu\text{s}$ is (approximately) 6 V

Ans. A, B, C

Sol. time constant of capacitor (τ) = $\frac{5RC}{8}$

Maximum charge on capacitor at steady state (q_0) = $5CV / 4$

Charge on capacitor at any time

$$q = \frac{5CV}{4} \left(1 - e^{-\frac{8t}{5RC}} \right) \dots (i)$$

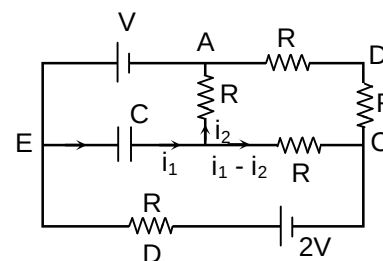
Current in the branch of capacitor at any time

$$i_1 = \frac{2V}{R} e^{-\frac{8t}{5RC}}$$

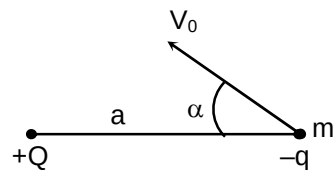
$$i_1 = 1.7 \text{ amp}$$

$$\text{Now, } V_A + i_2 R (i_1 - i_2) R = V_C$$

$$V_A - V_C = 2.52 \text{ volt}$$



7. A charged particle of charge $-q$ and mass m is projected with initial velocity $v_0 = \sqrt{\frac{qQ}{4\pi\epsilon_0 am}}$ from a distance a from the fixed charge $+Q$ at an angle of $\alpha = 60^\circ$ with the line joining the charges as shown in the figure. It is found that the particle moves on an elliptical orbit with the charge $+Q$ at one of its foci.



- (A) The length of the semi-major axis of the ellipse equals to a
 (B) The area of the ellipse orbit equals to $\frac{\sqrt{3}}{2} \pi a^2$

(C) The time period of one revolution will be $\sqrt{\frac{16\pi^3\epsilon_0 ma^3}{qQ}}$

(D) The time period of one revolution will be $\sqrt{\frac{8\pi^3\epsilon_0 ma^3}{qQ}}$

Ans. A, B, C

Sol. Total energy $E = \frac{1}{2}mv^2 - \frac{qQ}{4\pi\epsilon_0 a} = \frac{-qQ}{8\pi\epsilon_0 a}$

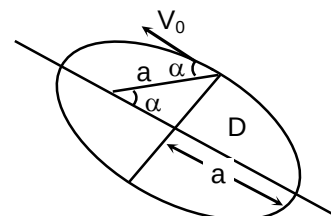
$$E = -\frac{qQ}{4\pi\epsilon_0 (2a)}$$

$\therefore$ The particle is moving on an ellipse of semi-major axis a

$$\text{and the time period will } T = 2\pi \sqrt{\frac{a^3}{\left(\frac{qQ}{4\pi\epsilon_0 m}\right)}} = \sqrt{\frac{16\pi^3 ma^3}{qQ}}$$

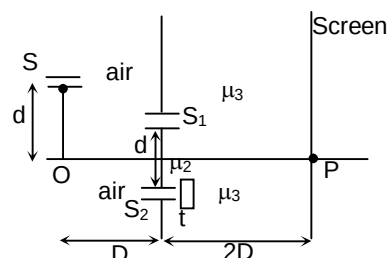
From geometry $b = a \sin \alpha$

$$\therefore \text{Area } A = \pi ab = \pi a^2 \sin \alpha = \frac{\sqrt{3}}{2} \pi a^2$$



8. Consider the situation shown in the figure. Two slits S_1 and S_2 are placed symmetrically about the line OP which is perpendicular to screen and bisector to line joining the slits. The space between screen and slits is filled with a liquid of refractive index μ_3 . A plate of thickness t and refractive index μ_2 is placed in front of one of the slit. A source S is placed above OP at a distance d in front of slit.

(Given that $D = 1\text{m}$, $d = 2\text{mm}$, $t = 6 \times 10^{-6}\text{m}$, $\mu_2 = 1.2$, $\mu_3 = 1.8$, Choose the correct alternatives)



- (A) Position of central maxima from point P is 2 mm.
 (B) Position of central maxima from point P is 1 mm.
 (C) If slab is removed, the central maxima shifts by a distance of 2 mm.
 (D) If slab is removed, the central maxima shifts by a distance of 1 mm.

Ans. A, C

Sol. Path difference = 0

$$\frac{d^2}{D} = \frac{yd}{2D} - \left(\frac{\mu_2}{\mu_3} - 1\right)t$$

$$y = 2\text{ mm}$$

$$\text{When slab is removed then path difference} = \frac{d^2}{D} - \frac{y_1 d}{2D} = 0, y_1 = 2d = 4\text{mm}$$

9. A particle of mass m is moving in a field where the potential energy is given by $U(x) = U_0(1 - \cos ax)$, where U_0 and a are constants and x is the displacement from mean position. Then (for small oscillations)

- (A) the time period is $T = 2\pi\sqrt{\frac{m}{aU_0}}$
- (B) the speed of particle is maximum at $x = 0$
- (C) the amplitude of oscillations is $\frac{\pi}{a}$
- (D) the time period is $T = 2\pi\sqrt{\frac{m}{a^2U_0}}$

Ans. B, C, D

Sol. $U = U_0(1 - \cos ax)$, $F = \frac{-dU}{dx} = -aU_0 \sin ax$

$$m \frac{d^2x}{dt^2} = -a^2U_0x \quad (\text{For small oscillations})$$

$$\Rightarrow T = 2\pi\sqrt{\frac{m}{a^2U_0}}$$

Potential energy is maximum at $x = \frac{\pi}{a}$

amplitude is $\frac{\pi}{a}$

Speed of particle is maximum at $x = 0$ ($\because$ PE is minimum)

10. Which of the following statement is correct regarding induced electric field (symbols have their usual meanings)?

- (A) Work done in moving a test charge in an induced electric field can be zero.
- (B) Induced electric field is non conservative in nature.
- (C) Induced electric lines of force form closed loops.
- (D) Induced e.m.f. in the loop is $\varepsilon = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\phi}{dt}$.

Ans. A, B, C, D

Sol. Induced electric field is a non-conservative field, therefore potential is not defined for such fields.

$$\text{e.m.f.} = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\phi}{dt}$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. A body executing SHM with time period T and amplitude A . When it is at $\frac{A}{2}$ from equilibrium position moving toward right. It receives an impulse which doubles its velocity without changing direction. Time taken by the block to reach the right extreme position is $t = T \cos^{-1}\left(\frac{K}{\sqrt{13}}\right)$. Find K .

Ans. 1

Sol. Velocity = $\omega\sqrt{A^2 - x^2}$
 $= \frac{2\sqrt{3}\pi A}{T}$ (doubles)
 $\frac{2\sqrt{3}\pi A}{T} = \frac{2\pi}{T} \sqrt{A^2 - x^2}$
 $A_1 = \frac{\sqrt{13}}{2} A$
 $\cos \theta = \frac{A/2}{A_1} = \frac{1}{\sqrt{13}} \quad \left(\theta = \cos^{-1}\left(\frac{1}{\sqrt{13}}\right) \right)$
 $\cos^{-1}\left(\frac{1}{\sqrt{13}}\right) = \omega t$
 $t = T \cos^{-1}\left(\frac{1}{\sqrt{13}}\right)$

12. A uniform elastic plank moves due to a force distributed uniformly over the end face. The cross-sectional area is A and Young's modulus Y . The strain produced in the direction of the force is $\frac{1}{X} \cdot \frac{F_0}{AY}$. Find X .

Ans. 2

Sol. The tension is not constant along the length of the plank. Let force at a distance x from one end be F .

Then $F = \frac{F_0 x}{l}$

Stress at any cross section = $\sigma = \frac{F}{A} = Y \times \text{strain} = \frac{F_0 x}{lA} \times \frac{\text{change in length of element } dx}{dx}$

Increase in length of element, $dx = \frac{F}{lAY} x dx \Rightarrow \Delta l = \frac{F_0}{lAY} \int_0^1 x dx$

Hence strain $\frac{\Delta l}{l} = \frac{1}{2} \cdot \frac{F_0}{AY}$

13. A radioactive sample contains two different types of radioactive nuclei; A-with half-life 5.0 days and B-with half-life 30.0 days. Initially the decay rate of the A type nuclei is 64 times that of B type of nuclei. Their decay rates will be equal when time is 9n days. Find the value of n.

Ans. 4

Sol. $\lambda_A = \frac{0.69}{5} \text{ day}^{-1}$ and $\lambda_B = \frac{0.69}{30} \text{ day}^{-1}$

Given $\left(\frac{dN_{OA}}{dt} \right) = 64 \left(\frac{dN_{OB}}{dt} \right)$

$\therefore \lambda_A N_{OA} = 64 \lambda_B N_{OB}$

At time t, $\frac{dN_A}{dt} = \frac{dN_B}{dt}$

$\therefore \lambda_A N_A = \lambda_B N_B$

Dividing equation (1) and (2) $\frac{N_{OA}}{N_A} = 64 \left(\frac{N_{OB}}{N_B} \right)$

But $N_A = N_{OA} \theta^{-2\lambda^1}$ and $N_B = N_{OB} \theta^{-2\lambda^1}$

$\therefore$ Equation (3) becomes $\theta^{\lambda^1} = 64 e^{2\lambda^1 t}$

$\therefore (\lambda_A - \lambda_B)t = 64$

$\therefore (\lambda_A - \lambda_B)t = \ln(64) = 2.303 \log_{10} 64$

$\therefore t = \frac{2.303 \log_{10} 64}{\left(\frac{0.693}{5} - \frac{0.693}{30} \right)} = 36 \text{ days}$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A cutting tool under microprocessor control has several forces acting on it. One force is $\vec{F} = -\alpha xy^2 \hat{j}$, a force in the negative y-direction whose magnitude depend on the position of the tool. The constant α is 2.50 N. Consider the displacement of the tool from the origin to the point $x = 3.00 \text{ m}$, $y = 3.00 \text{ m}$.

14. Calculate the magnitude of work done on the tool by $\vec{F}$. If this displacement is along the straight-line $y = x$ that connects these two points (in J)

Ans. 50. 60

Sol. $d\vec{r} = dx\hat{i} + dy\hat{j}$, $dw = \vec{F} \cdot d\vec{r} = -\alpha x y^2 dy$

On the path $x = y$, so $= -\alpha y^3 dy$,

$$w = \int_{y_1=0}^{y_2=3} -\alpha y^3 dy = -50.6$$

15. Calculate the work done on the tool by $\vec{F}$ of the tool is first moved out along the x-axis to the point $x = 3.00\text{m}$, $y = 0$ and then moved parallel to the y-axis to $x = 3.00\text{ m}$, $y = 3.00\text{ m}$ (in J)

Ans. 67.50

Sol. $dW = -\alpha xy^2 dy$ from $x = 0$ to $x = 3$ $dW = 0$
 $w = -2.50 \times 3 = 67.5\text{ J}$

Question Stem for Question Nos. 16 and 17

Question Stem

A steel tank has internal volume V_0 ($= 100\text{ litre}$). It contains half water (volume $= \frac{V_0}{2}$) and half kerosene oil at temperature $\theta_1 = 10^\circ\text{C}$

16. Calculate the mass of kerosene that flows out of the tank at temperature of $\theta_2 = 40^\circ\text{C}$. Coefficients of cubical expansion for different substances are: $\gamma_k = 10^{-3}^\circ\text{C}^{-1}$; $\gamma_w = 2 \times 10^{-4}^\circ\text{C}^{-1}$; $\gamma_{\text{steel}} = 1.2 \times 10^{-5}^\circ\text{C}^{-1}$. Density of kerosene at 10°C is $\rho_1 = 0.8\text{ kg/litre}$

Ans. 1.37

Sol. Volume of kerosene that spills out

$$\begin{aligned} \Delta V &= \Delta V_k + \Delta V_w - \Delta V_s \\ &= \frac{V_0}{2} \gamma_k \Delta\theta + \frac{V_0}{2} \gamma_w \Delta\theta - V_0 \gamma_s \Delta\theta = \frac{V_0 \Delta\theta}{2} (\gamma_k + \gamma_w - 2\gamma_s) \\ &= \frac{100 \times 30}{2} [10^{-3} + 0.2 \times 10^{-3} - 2 \times 0.012 \times 10^{-3}] \end{aligned}$$

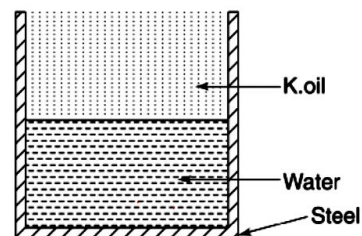
Density of kerosene at θ_2 is

$$\rho_2 = \frac{\rho_1}{1 + \gamma_k(\theta_2 - \theta_1)} = \frac{0.8}{1 + 10^{-3} \times 30} = \frac{0.8}{1.03} = 0.78\text{ kg/litre}$$

$\therefore$ Mass of K.oil that flows out is

$$\begin{aligned} \Delta m &= \Delta V \rho_2 = \frac{V_0 \rho_1 (\gamma_k + \gamma_w - 2\gamma_s)}{2 (1 + \gamma_k(\theta_2 - \theta_1))} (\theta_2 - \theta_1) \\ &= 0.78 \times 1.76 = 1.37\text{ kg} \end{aligned}$$

17. The height of water in the container at $\theta_1 = 10^\circ\text{C}$ is $H_1 = 1.0\text{ m}$. Find the height of water at $\theta_2 = 40^\circ\text{C}$.



Ans. 1.01

Sol. Volume of water at θ_2 is $V = \frac{V_0}{2}[1 + \gamma_w \Delta\theta]$ [where $\Delta\theta = \theta_2 - \theta_1$]

The area of cross section of the tank at θ_1 is $A_1 = \frac{V_0}{2H_1}$

At θ_2 cross section will be $A_2 = A_1[1 + \beta\Delta\theta] = A_1 \left[1 + \frac{2}{3}\gamma_s \Delta\theta \right] = \frac{V_0}{2H_1} \left[1 + \frac{2}{3}\gamma_s \Delta\theta \right]$

Height of water column is

$$H_2 = \frac{V}{A_2} = \frac{\frac{V_0}{2}[1 + \gamma_w \Delta\theta]}{\frac{V_0}{2H_1} \left[1 + \frac{2}{3}\gamma_s \Delta\theta \right]}$$

Question Stem for Question Nos. 18 and 19

Question Stem

A double pan window used for insulating a room thermally from outside consists of two glass sheets each of area 1 m^2 and thickness 0.01 m separated by 0.05 m thick stagnant air space. In the steady state, the room-glass interface and the glass-outdoor interface are at constant temperatures of 27°C and 0°C respectively. The thermal conductivity of glass is $0.8 \text{ Wm}^{-1}\text{K}^{-1}$ and of air $0.08 \text{ Wm}^{-1}\text{K}^{-1}$.

18. Calculate the temperature of the inner glass-air interface.

Ans. 26.50

19. Calculate the rate of flow of heat through the window pane.

Ans. 40.50

Sol. (for Q. 18 & 19)

Let T_2 and T_3 be the temperatures of the glass 1-air interface and air-glass 2 interface respectively

The equations of heat flow are $\frac{dQ_1}{dt} = \frac{dQ_2}{dt} = \frac{dQ_3}{dt}$

$$\Rightarrow \frac{dQ}{dt} = \frac{K_g A (T_1 - T_2)}{d_g} = \frac{K_a A (T_2 - T_3)}{d_a} = \frac{K_g A (T_3 - T_4)}{d_g} \quad \dots(A)$$

$$\Rightarrow \frac{dQ}{dt} = \frac{K_g A (300 - T_2)}{d_g} = \frac{K_a A (T_2 - T_3)}{d_a} = \frac{K_g A (T_3 - 273)}{d_g}$$

$$\Rightarrow 50 (300 - T_2) = T_2 - T_3 \quad \dots(i)$$

$$\text{And } 300 - T_2 = T_3 - 273 \quad \dots(ii)$$

Solve (i) and (ii) to get T_2 and T_3 . Put this in (A) to get $\frac{dQ}{dt}$

Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. If the elevation in boiling point of a solution of non-volatile, non-electrolytic and non-associating solute in solvent ($K_b = x \text{ K.Kg.mol}^{-1}$) is y K, then the depression in freezing point of solution of same concentration would be ($K_f = z \text{ K.kg.mol}^{-1}$)

(A) $\frac{2xz}{y}$

(B) $\frac{yz}{x}$

(C) $\frac{xz}{y}$

(D) $\frac{yz}{2x}$

Ans. B

Sol. $\Delta T_b = K_b \times m$

$\therefore m = \frac{y}{x} \dots(1)$

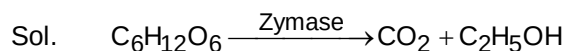
$\therefore \Delta T_f = K_f m \dots(2)$

$\Delta T_f = \left(\frac{y}{x}\right)z$

21. The function of zymase is to:

- (A) change starch into sugar
 (B) ferment glucose to alcohol and CO_2
 (C) change malt sugar into glucose
 (D) change starch into malt sugar and dextrin.

Ans. B



22. The density of a pure substance 'A' whose atoms are in cubic close pack arrangement is 1g/cc. If all the tetrahedral voids are occupied by 'B' atom, what is the density of resulting solid in g/cc. [Atomic mass (A) = 30 g/mol and atomic mass (B) = 50g/mol]

- (A) 3.33
(B) 4.33
(C) 2.33
(D) 5.33

Ans. B

Sol. Let volume of fcc unit cell = V

$$\rho_A = \frac{4 \times M_A}{N_A \cdot V}$$

$$\rho_B = \frac{8 \times M_B}{N_A \cdot V}$$

$$\frac{\rho_A}{\rho_B} = \frac{M_A}{2M_B} = \frac{30}{2 \times 50} = 0.3$$

$$\rho_B = \frac{\rho_A}{0.3} = 3.33 \text{ g/cc}$$

$$\text{Total density } \rho_A + \rho_B = 4.33 \text{ g/cc}$$

23. A reaction system in equilibrium according to reaction $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ in one litre vessel at a given temperature was found to be 0.12 mole each of SO_2 and SO_3 and 5 mole of O_2 . In another vessel of one litre contains 32 g of SO_2 at the same temperature. What mass of O_2 must be added to this vessel in order that at equilibrium 20% of SO_2 is oxidized to SO_3 ?

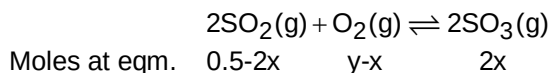
- (A) 0.4125g
(B) 11.6 g
(C) 1.6g
(D) 5.55 g

Ans. B

Sol. $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

$$K_c = \frac{(0.12)^2}{(0.12)^2 \times 5} = 0.2$$

Another vessel



$$\text{As per given } 2x = \frac{20}{100} \times 0.5 = 0.1$$

$$K_c = \frac{(0.1)^2}{(0.4)^2(y - 0.05)} = 0.20$$

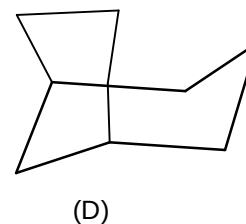
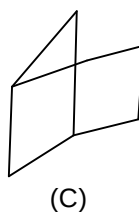
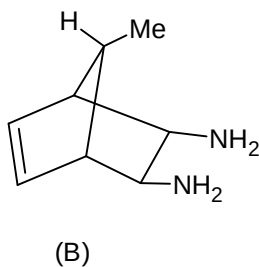
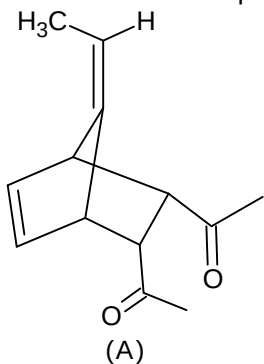
$$y = 0.3625 \text{ mole}$$

$$\therefore \text{mass of O}_2 \text{ added} = 11.6 \text{ g}$$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

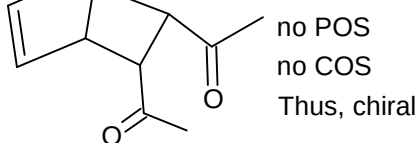
24. From the above compound (A), (B), (C) & (D) achiral compound is:



- (A) A
(B) B
(C) C
(D) D

Ans. B, C, D

Sol.  is perpendicular to the ring.



Plane of symmetry and centre of symmetry is not present in compound(A)

$\therefore$ It is chiral.

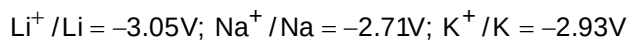
25. Select equations having endothermic step.

- (A) $\text{S}^-(\text{g}) \rightarrow \text{S}^{2-}(\text{g})$
(B) $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{NaCl}(\text{s})$
(C) $\text{N}(\text{g}) \rightarrow \text{N}^-(\text{g})$
(D) $\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g})$

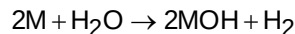
Ans. A, C, D

Sol. Factual

26. The standard electrode potentials of some of the alkali metals are



Alkali metals react with water according to the following reaction. (M = alkali metal)



It is found that Li reacts gently with water whereas potassium reacts violently with water.

Which of the following statement(s) is/are correct regarding the above two experimental facts?

- (A) ΔG values have nothing to do with the kinetics of a reaction
- (B) The metal having large standard oxidation potential will have lesser reactivity
- (C) K has a low melting point, and the heat of reaction is sufficient to make it melt or even vapourize. This leads to exponential increase in its surface area
- (D) The ionization potential of K is greater than that of Li.

Ans. A, C

Sol. Factual.

27. Amongst CuF_2 , CuCl_2 and CuBr_2 :

- (A) only CuF_2 is ionic
- (B) both CuCl_2 and CuBr_2 are covalent
- (C) CuF_2 and CuCl_2 are ionic but CuBr_2 is covalent
- (D) CuF_2 , CuCl_2 as well as CuBr_2 are ionic.

Ans. A, B

Sol. Factual

28. At 27°C , 1 mole of $\text{N}_2\text{O}_4(\text{g})$ placed in container of volume 8.21 lit, no dissociation will take place at this temperature. When gas is heated to 127°C , N_2O_4 is partially dissociated into NO_2 and final pressure is found to be 4.5 atm.

Select the correct statement.

- (A) No. of moles of NO_2 at 127°C is 0.5 moles
- (B) No. of moles of NO_2 at 127°C is 0.25 moles
- (C) Mole fraction N_2O_4 in gaseous mixture at 127°C is $\frac{7}{9}$
- (D) Ratio of the pressure of the system at 127°C will be $\frac{8}{9}$.

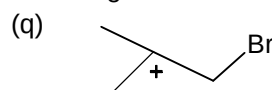
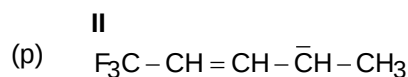
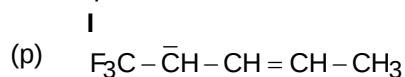
Ans. B, C, D

Sol. $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$
 1-x 2x

$$\text{Total moles} = 1 + x = \frac{4.5 \times 8.21}{0.0821 \times 300} = 1.5$$

$$x = 0.5$$

29. For the pairs of canonical forms written below what is/are true?



- (A) In both (p) and (q), I is more stable than II
 (B) In both (p) and (q), II is more stable than I
 (C) In (q), II is more stable than I
 (D) In (p), II is more stable than I

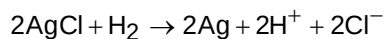
Ans. A

Sol. Factual

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. For the all reaction



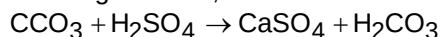
Temperature coefficient of emf of the cell is 0.02 VK^{-1} . ΔS° for the above reaction in kJ/K is (nearest integer).

Ans. 4

Sol. $\Delta S^\circ = nF \left[\frac{dE_{\text{cell}}}{dT} \right]_P$

$$= 2 \times 96500 \times 0.02 = 3.86 \text{ kJ/K}$$

31. 200 L of 600 ppm CaCO_3 dilute aq. Solution is mixed with 50L of 0.02 M H_2SO_4 solution using following reaction,



Calculate pH (excluding decimal part) if H_2CO_3 dissociates completely. [Given: $\log 4 = 0.6$]

Ans. 2

Sol. $n_{\text{H}_2\text{SO}_4} = 50 \times 0.02 = 1$

$$n_{\text{CaCO}_3} = \frac{600}{10^6} \times 200 \times 10^3 \times \frac{1}{100} = 1.2$$

$\therefore$ L.R. is H_2SO_4

$$[\text{H}_2\text{CO}_3] = \frac{1}{250} = 4 \times 10^{-3} \text{ M}$$

$$\text{pH} = -\log 2 \times 4 \times 10^{-3} = 3 - 3\log 2 = 3 - 0.903 = 2.1$$

32. 0.1 M HA is titrated against 0.1 M NaOH. Find pH at the end point. Dissociation constant for t acid HA is 5×10^{-6} and the degree of hydrolysis $h < 1$.

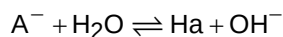
Ans. 9

Sol. $\text{HA} + \text{NaOH} \rightarrow \text{NaA} + \text{H}_2\text{O}$ at the end point, their equivalent amounts reacts together.
weak

$\therefore$ In the final solution.

$$[\text{NaA}] = \frac{0.1}{2} = 0.05 \text{ M}$$

As NaA is a salt of weak acid and strong base, it hydrolyses as,



$$\text{pH} = 7 + \frac{1}{2}[\text{pK}_a + \log c]$$

$$= 7 + \frac{1}{2}[-\log(5 \times 10^{-6}) + \log 0.05] = 9$$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

Electrolysis of cold sodium chloride solution produces NaOH and Cl_2 . Cl_2 produced, disproportionates into NaOH solution to give NaClO and NaCl. 10^3 L of 7.45% (w/w) solute of NaClO is produced if cell current is 9.65 ampere. Assume density of solution is 1 g/mL .

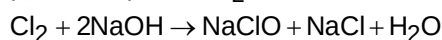
33. Find moles of Cl_2 required

Ans. 1000.00

34. How long will a cell operate to produce the above mentioned NaClO ($\times 10^7$)

Ans. 2.00

Sol. (Q33 & 34) $\text{Cl}^- + \text{Cl}_2 + 2\text{e}^-$



$$\text{Moles of NaClO required} = \frac{10^6 \times 1 \times 7.45 / 100}{7.45}$$

$$= 10^3 \text{ mol}$$

Moles of Cl_2 required = 10^3

Eq. Of Cl_2 required = $2 \times 10^7 = t$

Question Stem for Question Nos. 35 and 36

Question Stem

Standard Gibb's energy of reaction ($\Delta_r G^\circ$) at a certain temperature can be computed as $\Delta_r G^\circ = \Delta_r H^\circ$ and the change in the value of $\Delta_r H^\circ$ and $\Delta_r S^\circ$ for a reaction with temperature can be computed as follows:

$$\Delta_r H_{T_2}^\circ - \Delta_r H_{T_1}^\circ = \Delta_r C_p^\circ (T_2 - T_1)$$

$$\Delta_r S_{T_2}^\circ - \Delta_r S_{T_1}^\circ = \Delta_r C_p^\circ \ln \left(\frac{T_2}{T_1} \right)$$

$$\Delta_r G^\circ = \Delta_r H^\circ - T \cdot \Delta_r S^\circ$$

And by $\Delta_r G^\circ = -RT \ln K_{eq}$.

Consider the following reaction: $\text{CO(g)} + 2\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH(g)}$

Given: $\Delta_f H^\circ(\text{CH}_3\text{OH, g}) = -201 \text{ kJ/mol}$; $\Delta_f H^\circ(\text{CO, g}) = -114 \text{ kJ/mol}$

$S^\circ(\text{CH}_3\text{OH, g}) = 240 \text{ J/K} \cdot \text{mol}$; $S^\circ(\text{H}_2, \text{g}) = 29 \text{ J/K} \cdot \text{mol}^{-1}$

$S^\circ(\text{CO, g}) = 198 \text{ J/mol} \cdot \text{K}$; $C_{p,m}^\circ(\text{H}_2) = 28.8 \text{ J/mol} \cdot \text{K}$

$C_{p,m}^\circ(\text{CO}) = 29.4 \text{ J/mol} \cdot \text{K}$; $C_{p,m}^\circ(\text{CH}_3\text{OH}) = 44 \text{ J/mol} \cdot \text{K}$

And $\ln \left(\frac{320}{300} \right) = 0.06$, all data at 300 K.

35. $\Delta_r S^\circ$ at 320 K is:

Ans. -18.78

Sol.
$$\Delta_r S = -16 + \int_{300}^{320} \Delta_r C_p \frac{dT}{T}$$

$$\Delta_r S = -16 + (44 - 29.4 - 2 \times 28.8) \ln \frac{320}{300}$$

$$\Delta_r S = -16 + (-43) \ln \frac{320}{300} = -18.78$$

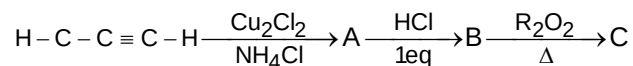
36. $\Delta_r H^\circ$ at 320 K is:

Ans. -87.86

Sol.
$$\Delta_r H = -87 + \int_{300}^{320} \Delta_r C_p dT$$

Question Stem for Question Nos. 37 and 38

Question Stem



Answer the following questions on the basis of above sequence of reactions:

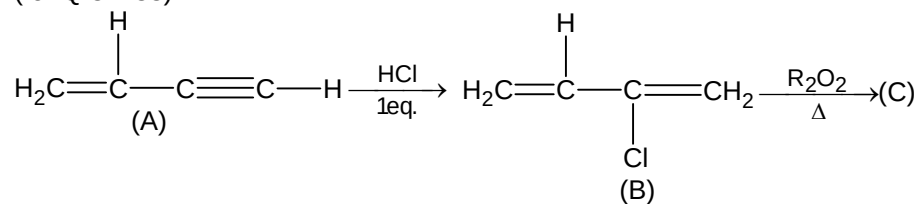
37. Degree of unsaturation in A is x. Find $\frac{x}{2}$.

Ans. 1.50

38. Ratio of number of carbon atom to number of H-atom in molecule B is

Ans. 0.80

Sol. (for Q.37–38)



C is artificial rubber (polymerization, free radical).

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. The equation of the line of latus rectum of the rectangular hyperbola $xy = c^2$ is

- (A) $x - y = \sqrt{2}c$
- (B) $x + y = 2\sqrt{2}c$
- (C) $x + y = \sqrt{2}c$
- (D) $x + y = 0$

Ans. B

Sol. $OA = c$ $A = (c, c)$

$AS = eOA$ ($\because e = \sqrt{2}$)

$AS = \sqrt{2}c$

$\therefore$ Co-ordinates of $S = (\sqrt{2}c, \sqrt{2}c)$

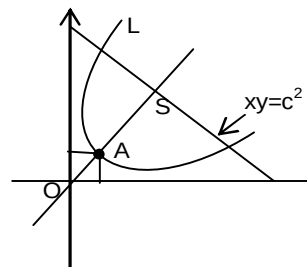
Equation of line through LL'

Slope of line = -1 and passing through $(\sqrt{2}c, \sqrt{2}c)$ is

$$(y - \sqrt{2}c) = -1(x - \sqrt{2}c)$$

$$y - \sqrt{2}c = -x + \sqrt{2}c$$

$$x + y = 2\sqrt{2}c.$$



40. The value of $\int (3x^{21} + 7x^9 + 21x^3)^{1/3} (x^{20} + x^8 + x^2) dx$ is

- (A) $\frac{1}{84} (3x^{28} + 7x^{12} + 21x^4) + c$
- (B) $\frac{1}{21} (3x^{20} + 7x^8 + 21x^2) + c$
- (C) $\frac{1}{80} (3x^{21} + 7x^9 + 21x^3) + c$
- (D) None of these

Ans. A

Sol. $\int (3x^{21} + 7x^9 + 21x^3)^{1/3} (x^{20} + x^8 + x^2) dx$

$$\text{put } 3x^{21} + 7x^9 + 21x^3 = t$$

$$63[x^{20} + x^8 + x^2]dx = dt$$

$$I_2 = \int \frac{1}{63} \cdot t^{\frac{1}{3}} dt = \frac{1}{63} \cdot \frac{t^{4/3}}{\frac{4}{3}} = \frac{1}{84} (3x^{21} + 7x^9 + 21x^3)^{4/3} + c$$

$$= \frac{1}{84} (3x^{28} + 7x^{12} + 21x^4) + c.$$

41. The sum to 'n' terms of the series whose n^{th} term is given by $T_n = (-1)^n \left(\frac{n^2 + n + 1}{n!} \right)$ is

- (A) $(-1)^n \frac{n+1}{n!} - 1$
 (B) $(-1)^n \frac{n}{(n+1)!} - 1$
 (C) $(-1)^n \frac{(n-1)}{(n+1)!} + 1$
 (D) None of these

Ans. A

Sol. $T_r = (-1)^r \frac{r^2 + r + 1}{r!} = (-1)^r \left[\frac{r}{(r-1)!} + \frac{r+1}{r!} \right]$

$$T_r = (-1)^r \frac{r+1}{r!} - (-1)^{r-1} \frac{r}{(r-1)!}$$

$$T_r = V_r - V_{r-1}$$

$$\sum_{r=1}^n T_r = V_n - V_0 = (-1)^n \frac{n+1}{n!} - 1$$

42. The co-ordinate of the foot of perpendicular drawn from the point (0, -2, 3) to the plane $2x - 3y + z = 5$ is

- (A) $\left(\frac{4}{7}, \frac{8}{7}, \frac{19}{7} \right)$
 (B) $\left(-\frac{4}{7}, \frac{8}{7}, \frac{19}{7} \right)$
 (C) $\left(-\frac{4}{7}, -\frac{8}{7}, \frac{19}{7} \right)$
 (D) None of these

Ans. C

Sol. Equation of line through $(0, -2, 3)$ and perpendicular to plane is

$$\frac{x-0}{2} = \frac{y+2}{-3} = \frac{z-3}{1} = k \text{ (say)}$$

Co-ordinates of a point on this line is $(2k, -2-3k, 3+k)$. This point when lying on the plane satisfies the equation of the plane. Hence $2(2k) - 3(-2-3k) + (3+k) = 5$.

$$\Rightarrow 4k + 6 + 9k + 3 + k = 5$$

$$\Rightarrow 14k = -4 \text{ or } k = \frac{-2}{7}.$$

$$\therefore \text{The point on the plane is } \left(\frac{-4}{7}, \frac{-8}{7}, \frac{19}{7} \right).$$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. Let $S_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^6} + \frac{32}{n^6} + \dots + \frac{1}{n} \right)$ and $T_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^6} + \frac{32}{n^6} + \dots + \frac{(n-1)^5}{n^6} \right)$, then which of the following is/ are true?

(A) $S_n \rightarrow \frac{1}{6}^+$

(B) $(S_n + T_n) < \frac{1}{3}$

(C) $(S_n + T_n) > \frac{1}{3}$

(D) $T_n \rightarrow \frac{1}{6}^-$

Ans. A, C, D

$$S_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^6} + \frac{32}{n^6} + \frac{243}{n^6} + \dots + \frac{1}{n} \right)$$

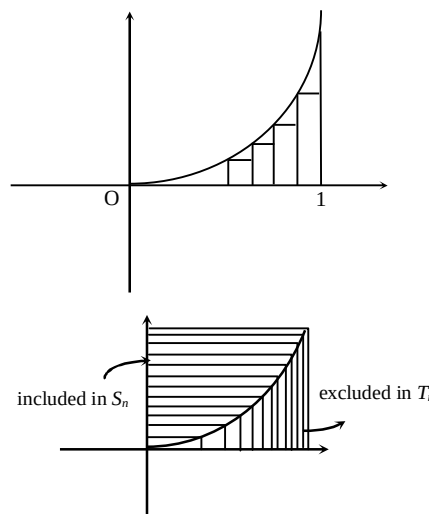
$$T_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^6} + \frac{32}{n^6} + \frac{243}{n^6} + \dots + \frac{(n-1)^5}{n^6} \right)$$

$$S_n = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r^5}{n^6} = \int_0^1 x^5 dx = \frac{x^6}{6} \Big|_0^1 = \frac{1}{6}$$

$$\Rightarrow S_n = \left(\frac{1}{6} \right)^+ ; T_n = \left(\frac{1}{6} \right)^-$$

But since x^5 is concave upward the area included in S_n for any two consecutive values of r is more than area excluded in T_n for same values of r .

$$\Rightarrow S_n - \frac{1}{6} > \frac{1}{6} - T_n \Rightarrow S_n + T_n > \frac{1}{3}$$



44. $z_1, z_2, \dots, z_n$ be non-zero complex numbers of equal modulus and satisfying the equation $z^n - |z|z^{n-1} + 1 = 0$, then

- (A) $\operatorname{Re}\left(\sum_{j=1}^n \sum_{k=1}^n \frac{z_j}{z_k}\right) = 0$
 (B) $\operatorname{Re}\left(\sum_{j=1}^n \sum_{k=1}^n \frac{z_j}{z_k}\right) = 1$
 (C) $\operatorname{Im}\left(\sum_{j=1}^n \sum_{k=1}^n \frac{z_j}{z_k}\right) = 0$
 (D) $\operatorname{Im}\left(\sum_{j=1}^n \sum_{k=1}^n \frac{z_j}{z_k}\right)$

Ans. B, C

Sol. Let $S = \sum_{j=1}^n \sum_{k=1}^n \frac{z_j}{z_k} = (z_1 + z_2 + \dots + z_n) \left(\frac{1}{z_1} + \frac{1}{z_2} + \dots + \frac{1}{z_n} \right)$

If $|z_1| = |z_2| = \dots = |z_n| = r$

$$\Rightarrow \frac{1}{z_k} = \frac{\bar{z}_k}{r^2}$$

$$\Rightarrow S = \frac{1}{r^2} (z_1 + z_2 + \dots + z_n) (\bar{z}_1 + \bar{z}_2 + \dots + \bar{z}_n)$$

$$= \frac{|z_1 + z_2 + \dots + z_n|^2}{r^2}$$

But $z_1 + z_2 + \dots + z_n = |z| = r$

$$\Rightarrow S = \frac{r^2}{r^2} = 1$$

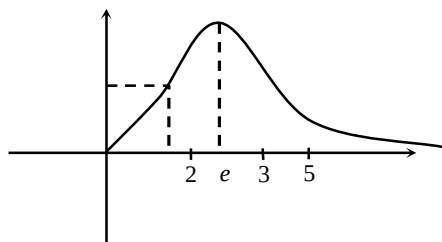
45. If $\log_2 \left(\log_{\frac{1}{2}} (\log_2(x)) \right) = \log_3 \left(\log_{\frac{1}{3}} (\log_3(y)) \right) = \log_5 \left(\log_{\frac{1}{5}} (\log_5(z)) \right) = 0$ for positive x, y and z , then which of the following is/ are NOT true?

- (A) $z < x < y$
 (B) $x < y < z$
 (C) $y < z < x$
 (D) $z < y < x$

Ans. B, C, D

Sol. Solving we get $x = 2^{1/2}$, $y = 3^{1/3}$, $z = 5^{1/5}$

Using graph of $x^{1/x}$



$$\Rightarrow 3^{1/3} > 5^{1/5}$$

$$\text{Also } 2^{1/2} < 3^{1/3} \quad \left| \quad 2^{1/2} > 5^{1/5} \right.$$

$$\text{as } 2^3 < 3^2 \quad \left| \quad 2^5 > 5^2 \right.$$

$$\Rightarrow y > x > z$$

Hence (B), (C) & (D) are NOT true

46. A drawer contains red and black balls. When two balls are drawn at random, the probability that they both are red is $\frac{1}{2}$. The number of balls in the drawer can be

- (A) 21
- (B) 11
- (C) 4
- (D) 3

Ans. A, C

Sol. Let the drawer contains p balls of which 'm' are red.

$$\text{Probability of drawing two red balls at random is } \frac{{}^m C_2}{{}^p C_2} = \frac{1}{2}$$

$$\Rightarrow 2m(m-1) = p(p-1)$$

$$\Rightarrow 2m^2 - 2m - p^2 + p = 0$$

$$\Rightarrow m = \frac{2 \pm \sqrt{4 - 8(p - p^2)}}{4} = \frac{1 \pm \sqrt{1 - 2p + 2p^2}}{2}$$

$$\Rightarrow 1 - 2p + 2p^2 \text{ should be an odd perfect square.}$$

$$\text{i.e., } p = 21, 4 \text{ but } p \neq 3$$

when 3 balls out of 4 are red.

15 balls out of 21 are red.

47. If a function satisfies $(x-y)f(x+y) - (x+y)f(x-y) = 2(x^2y - y^3) \forall x, y \in \mathbb{R}$ and $f(1) = 2$, then

- (A) $f(x)$ must be polynomial function
- (B) $f(3) = 12$
- (C) $f(0) = 0$
- (D) $f(x)$ may not be differentiable

Ans. A, B, C

Sol. $(x-y)f(x+y) - (x+y)f(x-y) = 2y(x-y(x+y))$

Let $x-y = u$; $x+y = v$

$uf(v) - vf(u) = 2uv(v-u)$

$\frac{f(v)}{v} - \frac{f(u)}{u} = v - u$

$\left(\frac{f(v)}{v} - v\right) = \left(\frac{f(u)}{u} - u\right) = \text{constant}$

Let $\frac{f(x)}{x} - x = \lambda$

$\Rightarrow f(x) = (\lambda x + x^2)$

$f(1) = 2$

$\lambda + 1 = 2 \Rightarrow \lambda = 1$

$f(x) = x^2 + x$

48. In a $\triangle ABC$ medians AD and BE are perpendicular to each other, then

(A) $\frac{1}{2} < \frac{a}{b} < 2$

(B) $\frac{1}{2} < \frac{c}{a} < 2$

(C) $\frac{1}{2} < \frac{a}{c} < 2$

(D) $\frac{1}{2} < \frac{b}{a} < 2$

Ans. A, D

Sol. $AG = \frac{1}{3}\sqrt{2b^2 + 2c^2 - a^2}$ and $BG = \frac{1}{3}\sqrt{2a^2 + 2c^2 - b^2}$

$AG^2 + BG^2 = c^2$

$\Rightarrow (2b^2 + 2c^2 - a^2) + (2a^2 + 2c^2 - b^2) = 9c^2$

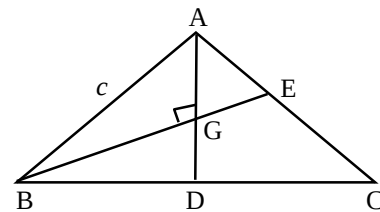
$\Rightarrow a^2 + b^2 = 5c^2 \quad \dots(i)$

and for $\triangle ABC$ exist

$a - b < c$ or $b - a < c$

$\Rightarrow (a-b)^2 < c^2 \Rightarrow a^2 + b^2 - 2ab < \frac{a^2 + b^2}{5} \Rightarrow 2a^2 + 2b^2 - 5ab < 0$

$\Rightarrow 2\left(\frac{a}{b}\right)^2 - 5\left(\frac{a}{b}\right) + 2 < 0 \Rightarrow \frac{a}{b} = \frac{5 \pm \sqrt{25-16}}{4} = \frac{5 \pm 3}{4} = 2, \frac{1}{2}$



$$\Rightarrow \frac{1}{2} < \frac{a}{b} < 2$$

$$\text{Similarly, } \frac{1}{2} < \frac{b}{a} < 2$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. The locus of the vertices of the family of parabolas $y = \frac{a^3}{3}x^2 + \frac{a^2}{2}x - 2a$ is $64xy = 21k$, where k is equal to

Ans. 5

Sol. Given equation is

$$y = \frac{a^3 x^2}{3} + \frac{a^2 x}{2} - 2a$$

$$\Rightarrow y + 2a = \frac{a^3}{3} \left\{ x^2 + \frac{3}{2a}x \right\}$$

$$\Rightarrow y + 2a = \frac{a^3}{3} \left\{ \left(x + \frac{3}{4a} \right)^2 - \frac{9}{16a^2} \right\}$$

$$\Rightarrow \frac{3}{a^3} (y + 2a) + \frac{9}{16a^2} = \left(x + \frac{3}{4a} \right)^2$$

$$\Rightarrow \frac{3}{a^3} \left\{ y + 2a + \frac{3}{16}a \right\} = \left(x + \frac{3}{4a} \right)^2$$

$$\Rightarrow \frac{3}{a^3} \left(y + \frac{35}{16}a \right) = \left(x + \frac{3}{4a} \right)^2$$

Which is of the form $X^2 = 4AY$.

Vertex is given by

$$X = 0, Y = 0$$

$$\Rightarrow x + \frac{3}{4a} = 0$$

$$\text{and } y + \frac{35}{16}a = 0$$

$$\Rightarrow x = -\frac{3}{4a} \text{ and } y = -\frac{35}{16}a \Rightarrow xy = \frac{105}{64}$$

$$\Rightarrow \frac{105}{64} = \frac{21}{64}k \Rightarrow k = 5.$$

50. Tangents are drawn from the point $(4, 3)$ to the circle $x^2 + y^2 = 9$. Then the area of the triangle formed by these tangents and the chord of contact is $\frac{64\Delta}{25}$ square units where Δ is equal to

Ans. 3

Sol. The equation of the chord of contact of the two tangents that can be drawn from $P(4,3)$ to the circle $x^2 + y^2 = 9$ is ($S_1 = 0$)

$$4x + 3y - 9 = 0 \quad \dots(1)$$

For area of ΔPAB , we need

$$|AB| \text{ and } |MP|$$

$$\text{Here, } |OP| = \sqrt{(4-0)^2 + (3-0)^2} = 5$$

Also, $|OM| =$ distance of $O(0,0)$ from line (1)

$$= \frac{|0+0-9|}{\sqrt{4^2+3^2}} = \frac{9}{5}$$

$$\begin{aligned} \therefore |AB| &= 2\sqrt{(\text{radius})^2 - OM^2} \\ &= 2\sqrt{3^2 - \left(\frac{9}{5}\right)^2} = 2\sqrt{\frac{144}{25}} = \frac{24}{5} \end{aligned}$$

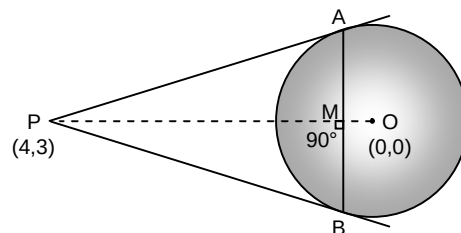
$$\text{and } |MP| = |OP| - |OM|$$

$$= 5 - \frac{9}{5} = \frac{16}{5}$$

$$\text{Hence area } \Delta PAB = \frac{1}{2} |AB| |MP|$$

$$= \frac{1}{2} \left(\frac{24}{5}\right) \left(\frac{16}{5}\right) = \frac{192}{25} \text{ square units}$$

$$\Rightarrow \frac{64\Delta}{25} = \frac{192}{25} \Rightarrow \Delta = 3.$$



51. A straight line L with negative slope passes through the point $(8, 2)$ and cuts the positive co-ordinate axes at point P and Q . As L varies, the absolute minimum value of $OP + OQ$ is (O is origin)

Ans. 18

Sol. The equation of line L be $y - 2 = m(x - 8)$, $m < 0$ co-ordinates of P and Q are

$$P\left(8 - \frac{2}{m}, 0\right) \text{ and } Q(0, 2 - 8m).$$

$$\text{So, } OP + OQ = 8 - \frac{2}{m} + 2 - 8m$$

$$= 10 + \frac{2}{(-m)} + 8(-m) \geq 10 + 2\sqrt{\frac{2}{(-m)} \times 8(-m)} \geq 18$$

So, absolute minimum value of

$$OP + OQ = 18$$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S. Now answer the following questions

52. The ratio of the areas of the triangles PQS and PQR is

Ans. 0.25

Sol. $P \equiv (1, 2\sqrt{2})$ $Q = (1, -2\sqrt{2})$

Equation of PR : $x + 2\sqrt{2}y = 9$

$R \equiv (9, 0)$

Equation of PS : $2\sqrt{2}y = 4(x+1)$

$S \equiv (-1, 0)$

$$\text{Area } \frac{PQS}{PQR} = \frac{\frac{1}{2} \times 2 \times 4\sqrt{2}}{\frac{1}{2} \times 8 \times 4\sqrt{2}} = \frac{1}{4}$$

53. The radius of the circumcircle of the triangle PQR is

Ans. 4.50

Sol. Diameter of circum circle of ΔPQR will be $OR = 9$
Where O is origin

Question Stem for Question Nos. 54 and 55

Question Stem

If the locus of the circumcentre of a variable triangle having sides y-axis, $y = 2$ and $\ell x + my = 1$, where (ℓ, m) lies on the parabola $y^2 = 4ax$ is curve C, then

54. Coordinates of the vertex of this curve C is $(-2a, \beta)$ then the value of $\beta^2 + \beta + 101$ is

Ans. 104.75

55. The curve C is symmetric about the line $y = k$ then value of $|2021k|$

Ans. 3031.50

Sol. $\therefore C \equiv \left(0, \frac{1}{m}\right)$

$B \equiv \left(\frac{1-2m}{\ell}, 2\right), A(0, 2)$

Let (h, k) be the circumcentre of $\triangle ABC$

$\therefore h = \frac{1-2m}{2\ell} : k = \frac{1+2m}{2m}$

$\therefore 2h = \frac{1-2m}{\ell} : k = 1 + \frac{1}{2m}$

$\therefore m = \frac{1}{2k-2} : \ell = \frac{k-2}{2h(k-1)}$

$\therefore (\ell, m) \text{ lies on } y^2 = 4ax$

$\therefore m^2 = 4a\ell$

$\Rightarrow \left(\frac{1}{2k-2}\right)^2 = 4a \left\{ \frac{k-2}{2h(k-1)} \right\}$

$h = 8a(k^2 - 3k + 2)$

$\therefore$ locus of (h, k) is

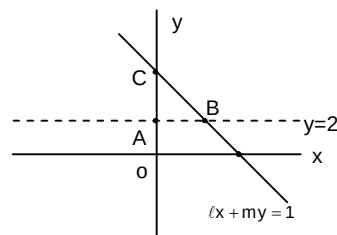
$x = 8a(y^2 - 3y + 2)$

$\Rightarrow \left(y - \frac{3}{2}\right)^2 = \frac{1}{8a}(x + 2a)$

$\therefore$ vertex is $\left(-2a, \frac{3}{2}\right)$

$\therefore$ Length of smallest focal chord = length of latus rectum = $\frac{1}{8a}$

From the equation of curve C it is clear that it is symmetric about line $y = \frac{3}{2}$.



Question Stem for Question Nos. 56 and 57

Question Stem

Consider a general equation of degree 2, as $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$

56. The value of ' λ ' for which the line represents a pair of straight lines is

Ans. 2.00

57. For the value of λ obtained in above question, if $L_1 = 0$ and $L_2 = 0$ are the lines denoted by the given line pair then the absolute value of the sum of abscissa and ordinate of their point of intersection is

Ans. 13.50

Sol. (for Q. 56 & 57)

$$a = \lambda; \quad h = -5; \quad b = 12; \quad g = (5/2); \quad f = -8, \quad c = -3$$

$$\lambda(12)(-3) + 2(-8)(5/2)(-5) - \lambda(64) - (25/4) \cdot 12 + 3 \cdot 25 = 0$$

$$-36\lambda + 200 - 64\lambda - 75 + 75 = 0 \Rightarrow 100\lambda = 200 \Rightarrow \lambda = 2 \text{ Ans.}$$

$$(ii) 2x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$$

consider the homogeneous part

$$2x^2 - 10xy + 12y^2$$

$$2x^2 - 6xy - 4xy + 12y^2$$

$$\text{or } 2x(x - 3y) - 4y(x - 3y)$$

$$\text{or } (x - 3y)(x - 2y)$$

$$\therefore 2x^2 - 10xy + 12y^2 + 5x - 16y - 3 \equiv (2x - 6y + A)(x - 2y + B)$$

$$\text{solving } A = -1; \quad B = 3$$

$$\text{hence lines are } 2x - 6y - 1 = 0 \quad \text{and} \quad x - 2y + 3 = 0$$

$$\text{solving intersection point } (-10, -7/2)$$

$$\therefore \text{product} = 35$$